

UCI: Division of Continuing Education
 Accelerated Certificate Program (ACP)
 Data Science
 Introduction to Python Programming
 I&C SCI_X426.58:
 Homework#7

Date Given: April 13, 2023

Due Date: April 17, 2023

There are 10 problems in this homework assignment. Write a program in Python to solve these problems. Use the '**Pandas**' package to write your Python code.

Use the data file (top_movies_by_title.csv) for all problems of this assignment. This file contains information about 200 Hollywood movies. This file is uploaded on Canvas available for download adjacent to the homework assignment#7 (HW07.pdf) file. The first few rows of this file are displayed in the following figure.

	A	B	C	D	E
1	Title	Studio	Gross	Gross (Adjusted)	Year
2	101 Dalmatians	Disney	144880014	869280100	1961
3	2001: A Space Odyssey	MGM	56954992	377027700	1968
4	9 to 5	Fox	103290500	334062200	1980
5	A Star Is Born (1954)	Warner Bros	80000000	326760600	1976
6	Air Force One	Sony	172956409	327752300	1997
7	Airport	Universal	100489151	575168200	1970
8	Aladdin	Buena Vista (Walt Disney)	217350219	456248400	1992
9	Alice in Wonderland (2010)	Buena Vista (Walt Disney)	334191110	365718600	2010
10	American Graffiti	Universal	115000000	571714300	1973
11	American Sniper	Warner Bros	350126372	374796000	2014
12	An Officer and a Gentleman	Paramount	129795554	379814600	1982

The 'Gross (Adjusted)' amount is the inflation adjusted gross amount. For example, '101 Dalmatians' movie did \$144 Million in 1961, it is equivalent to \$869 Million in today's dollars.

1. Read the 'top_movies_by_title.csv' file into a Pandas' data frame and name it 'table1'.

2. Select 'Title' and 'Year' of the 'table1', into another table and name is 'table2'.

You output should look as follows.

	Title	Year
0	101 Dalmatians	1961
1	2001: A Space Odyssey	1968
2	9 to 5	1980
3	A Star Is Born (1976)	1976
4	Air Force One	1997
...	...	...
195	Up	2009
196	West Side Story	1961
197	What's Up, Doc?	1972
198	Who Framed Roger Rabbit	1988
199	Young Frankenstein	1974

200 rows × 2 columns

3. Select only the movies from 'table1' where the gross revenues generated by the movie exceeds 100 million.

You output should look as follows.

	Title	Studio	Gross	Gross (Adjusted)	Year
0	101 Dalmatians	Disney	144880014	869280100	1961
2	9 to 5	Fox	103290500	334062200	1980
4	Air Force One	Sony	172956409	327752300	1997
5	Airport	Universal	100489151	575168200	1970
6	Aladdin	Buena Vista (Disney)	217350219	456248400	1992
...	...	...	...	...	...
192	Transformers: Dark of the Moon	Paramount/Dreamworks	352390543	385069700	2011
193	Transformers: Revenge of the Fallen	Paramount/Dreamworks	402111870	468938100	2009
194	Twister	Warner Bros.	241721524	475786700	1996
195	Up	Buena Vista (Disney)	293004164	341681900	2009
198	Who Framed Roger Rabbit	Buena Vista (Disney)	156452370	331176600	1988

160 rows × 5 columns

4. Select only the movies from 'table1' where the gross revenues generated by the movie is between 100 and 200 million. Sort the output data frame by 'Gross' revenues in descending order. Display only the top 10 entries.

You output should look as follows.

	Title	Studio	Gross	Gross (Adjusted)	Year
51	Gone with the Wind	MGM	198676459	1757788200	1939
69	Indiana Jones and the Last Crusade	Paramount	197171806	429923500	1989
188	Toy Story	Buena Vista (Disney)	191796233	381654400	1995
52	Grease	Paramount	188755690	669632000	1978
129	Snow White and the Seven Dwarfs	Disney	184925486	948300000	1937
35	Dances with Wolves	Orion	184208848	380183900	1990
20	Batman Forever	Warner Bros.	184031112	368062200	1995
154	The Fugitive	Warner Bros.	183875760	386256600	1993
83	Liar Liar	Universal	181410615	343850200	1997
91	Mission: Impossible	Paramount	180981856	356231300	1996

5. Select movies where the word 'the' appears in the title of the movie. Show the first 10 movies.

You output should look as follows.

	Title	Studio	Gross	Gross (Adjusted)	Year
13	Around the World in 80 Days	UA	42000000	562153800	1956
17	Back to the Future	Universal	210609762	513740700	1985
22	Beauty and the Beast	Buena Vista (Disney)	218967620	394664300	1991
28	Butch Cassidy and the Sundance Kid	Fox	102308889	613853300	1969
33	Close Encounters of the Third Kind	Columbia	132088635	483507200	1977
38	Duel in the Sun	Selz.	20408163	443877500	1946
51	Gone with the Wind	MGM	198676459	1757788200	1939
54	Guardians of the Galaxy	Buena Vista (Disney)	333176600	358244800	2014
56	Harry Potter and the Chamber of Secrets	Warner Bros.	261988482	390768100	2002
57	Harry Potter and the Deathly Hallows Part 2	Warner Bros.	381011219	417512200	2011

6. Sort the movies in 'table1' in descending order (highest value first) by their 'Gross' revenues. Select top 3 movies made the most amount of money based on 'Gross' column? Select top 3 movies made the most amount of money based on 'Gross (Adjusted)' column?

You output should look as follows.

	Title	Studio	Gross	Gross (Adjusted)	Year
137	Star Wars: The Force Awakens	Buena Vista (Disney)	906723418	906723400	2015
15	Avatar	Fox	760507625	846120800	2009
185	Titanic	Paramount	658672302	1178627900	1997

You output should look as follows.

	Title	Studio	Gross	Gross (Adjusted)	Year
51	Gone with the Wind	MGM	198676459	1757788200	1939
133	Star Wars	Fox	460998007	1549640500	1977
176	The Sound of Music	Fox	158671368	1239013800	1965

7. Count the number of movies from each studio. How many movies made by 'Columbia' and 'Fox' studios?

You output should look as follows.

	count
Studio	
Warner Bros.	29
Buena Vista (Disney)	29
Fox	26
Paramount	25
Universal	22
Disney	11
Columbia	10
MGM	7
UA	6
Sony	6

8. Which is the highest grossing movie from 'Columbia' studio?

You output should look as follows.

	Title	Studio	Gross	Gross (Adjusted)	Year	count
49	Ghostbusters	Columbia	242212467	619211400	1984	1

9. Compute the ratio between 'Gross (Adjusted)' and 'Gross' for all movies. Next sort the table by the ratio in descending order. Display the top 10 ratios.

You output should look as follows.

	Title	Studio	Gross	Gross (Adjusted)	Year	count	Ratio
153	The Four Horsemen of the Apocalypse	MPC	9183673	399489800	1921	1	43.500003
143	The Bells of St. Mary's	RKO	21333333	545882400	1945	1	25.588238
121	Sergeant York	Warner Bros.	16361885	418671800	1941	1	25.588238
38	Duel in the Sun	Selz.	20408163	443877500	1946	1	21.749998
144	The Best Years of Our Lives	RKO	23650000	478500000	1946	1	20.232558
65	House of Wax	Warner Bros.	23750000	439627700	1953	1	18.510640
146	The Caine Mutiny	Columbia	21750000	386173500	1954	1	17.755103
178	The Ten Commandments	Paramount	65500000	1139700000	1956	1	17.400000
145	The Bridge on the River Kwai	Columbia	27200000	473280000	1957	1	17.400000
173	The Robe	Fox	36000000	569454500	1953	1	15.818181

10. Group all the movies by the year in which they were released. In which year has the maximum number of movies released?

You output should look as follows.

	count
Year	
2015	6
2009	6
2004	6
2002	6
2012	5
2007	5
2003	5
2001	5
1997	5
1982	5